Intelligent Spreadsheet Analytics & Automation Platform

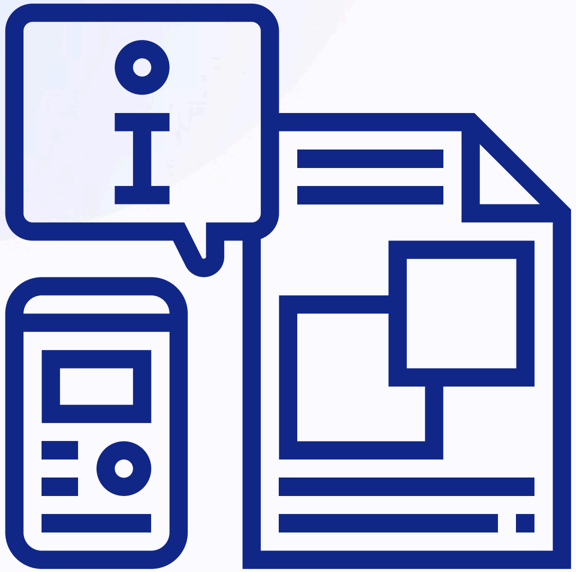
Internship Work Report — Delhi Technological University (DTU)

ANUGYA SAXENA

3-MONTH INTERNSHIP | COMPUTER SCIENCE ENGINEERING | ON-SITE

Internship Details

Anugya Saxena



Institution

Delhi Technological University (DTU), a premier engineering institution in India known for its rigorous Computer Science and Engineering programs.

Program

Bachelor of Technology (B.Tech) — Computer Science Engineering (CSE), focusing on software systems, data engineering, and intelligent automation.

Duration & Mode

3 months, conducted offline and on-site — enabling direct collaboration, real-time feedback, and immersion in a professional development environment.

Problem Statement

Organizations and offices frequently manage large volumes of spreadsheet-based data for project management, resource tracking, reporting, and operational workflows. Traditional spreadsheet management requires significant manual effort for searching, filtering, updating, and generating reports — resulting in inefficiencies and increased chances of human error.

The Core Challenge

Manual spreadsheet operations consume valuable time and introduce errors. Non-technical users struggle with complex formulas, conditional formatting, and data filtering — creating bottlenecks in operational workflows and delaying critical decision-making processes across teams.

The Proposed Solution

This project addresses the challenge by developing an **intelligent spreadsheet analytics and automation platform** that enables users to interact with spreadsheet data through natural language queries while providing automated reporting, visualization, and full CRUD (Create, Read, Update, Delete) operations — all without requiring advanced technical expertise.

Manual Effort

Repetitive searching, filtering, and updating tasks consume significant staff hours across operational teams.

Human Error

Manual data handling introduces inaccuracies that propagate through reports and downstream decisions.

Accessibility Gap

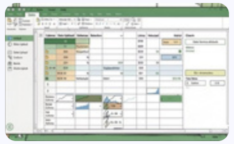
Non-technical users are excluded from meaningful data analysis due to complexity of traditional tools.

Slow Reporting

Report generation is time-consuming and requires specialized knowledge of formulas and pivot tables.

Work Completed

Overthecourse oftheinternship,a comprehensive suite of features was designed and implemented, forming a fully functional GUI-based spreadsheetmanagement platform. Theworkspans the full stack — from intelligent data ingestion and schema detection to natural language processing and automated visualization.



Core Platform & Data Ingestion

Developed a GUI-based spreadsheet management platform with Excel upload functionality and intelligent schema detection that automatically identifies data types, column relationships, and structural patterns within uploaded files.

CRUD Operations & Query Handling

Implemented full CRUD (Create, Read, Update, Delete) operations for spreadsheet management alongside dynamic query handling and search functionality that enables rapid data retrieval across large datasets.

Reporting & Visualization

Built automated report generation and visualization modules that transform raw spreadsheet data into meaningful charts, summaries, and structured reports — eliminating manual report preparation.

Natural Language & LLM Integration

Integrated Natural Language Query Processing capabilities and a local LLM (Large Language Model) integration framework, enabling users to ask questions about their data in plain, conversational English.

Dynamic filtering and conditional analytics support were also developed, allowing users to apply complex logical conditions to their data without writing formulas manually.

Project Impact

The platform delivers measurable improvements across operational efficiency, accessibility, and analytical capability. By automating repetitive spreadsheet tasks and introducing intelligent interaction layers, the system addresses pain points experienced by both technical and non-technical users in organizational settings.

3

Months of Development

A fully functional platform delivered within a structured 3-month internship timeline.

8+

Core Features Built

From Excel upload to LLM integration, eight major feature sets were designed and implemented.

0

Formula Expertise Required

Non-technical users can perform complex analytics without knowledge of spreadsheet formulas.

Operational Benefits

- Reduces manual spreadsheet operations and repetitive tasks, freeing staff for higher-value work
- Improves efficiency in handling large datasets that would otherwise require significant manual effort
- Enables faster analysis and report generation, accelerating decision-making cycles
- Increases productivity through intelligent automation of routine data workflows

Strategic Benefits

- Provides accessible analytics for non-technical users, democratizing data access across teams
- Reduces dependency on complex formulas and manual filtering, lowering the barrier to entry
- Creates a foundation for scalable analytics systems within operational environments
- Establishes a reusable architecture that can be extended for enterprise-wide deployment

Expected Outcomes

The platform is designed to deliver a set of concrete, measurable outcomes that directly address the inefficiencies identified in the problem statement. These outcomes span user experience, operational efficiency, and analytical capability.

→ Intelligent Spreadsheet Interaction

Users will be able to interact with spreadsheet data using conversational queries, eliminating the need to learn complex formula syntax or navigate intricate menu structures. The local LLM integration interprets natural language and translates it into precise data operations.

→ Faster Report Generation & Analytics

Automated report generation modules will significantly reduce the time required to produce structured reports, charts, and summaries. What previously required hours of manual preparation will be accomplished in seconds through automated pipelines.

→ Reduced Operational Effort

By automating repetitive spreadsheet management tasks — including data entry validation, filtering, sorting, and conditional formatting — the platform will substantially reduce the operational burden on staff, allowing them to focus on interpretation and decision-making.

→ Improved Decision-Making

Automated insights derived from the data will support faster and more informed decision-making. The platform surfaces patterns, anomalies, and trends that might otherwise go unnoticed in manual analysis workflows.

→ Reusable Analytics Workflows

The system will enable the creation of reusable analytics workflows that can be applied consistently across different datasets and departments, promoting standardization and reducing the learning curve for new users.

Future Development Roadmap

The platform has been architected with extensibility in mind. The following roadmap outlines planned enhancements that will evolve the system from a functional prototype into a production-ready, enterprise-grade analytics solution capable of supporting complex organizational workflows.

1

Phase 1 – Enhanced NLP

Improve Natural Language Query understanding capabilities to handle more complex, multi-part queries and contextual follow-up questions within a conversation.

2

Phase 2 – Multi-Sheet Analytics

Add support for multi-sheet analytics, enabling cross-sheet joins, aggregations, and relational queries across workbook tabs.

3

Phase 3 – Advanced Dashboards

Develop advanced dashboard visualization modules with interactive charts, KPIs, and drill-down capabilities for executive-level reporting.

4

Phase 4 – Export & Integration

Integrate PDF and automated report export systems, alongside database connectivity and backend storage for persistent data management.

5

Phase 5 – Cloud & Collaboration

Add cloud deployment capabilities, extend support for larger datasets and enterprise workflows, and enable collaborative multi-user functionality.

Each phase builds upon the existing architecture, ensuring backward compatibility while progressively expanding the platform's capabilities to meet enterprise-scale requirements.

Integration Within Office Environments

Potential Integration Areas

Project Management Tracking

Automate status updates, milestone tracking, and resource allocation reports across active projects.

Resource Allocation Monitoring

Track utilization, availability, and distribution of human and material resources in real time.

Administrative Reporting

Generate structured administrative reports automatically, reducing the burden on administrative staff.

Operational Data Analytics

Enable non-technical staff to perform operational analytics through conversational queries.

Spreadsheet Automation Workflows

Replace manual, repetitive spreadsheet tasks with automated workflows triggered by rules or queries.

Decision Support Systems

Provide data-driven insights that support faster, more confident decision-making at all organizational levels.

> The platform can serve as an intelligent layer above existing spreadsheet-based workflows , enhancing capability without disrupting established processes or requiring costly infrastructure overhauls.